From Detection to Interpretation

Transformer-Based Cross-Lingual Approaches to Religious Hate Speech

A *who–how–why* framework for Digital Religion research
Case: Anti-Christian discourse in Chinese, Japanese, and English

Zhidian Huang

University of Zurich · Department of Computational Linguistics

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"How computational methods can reveal cross-cultural structures of religious hostility in digital spaces."

Why this talk belongs at Digital Religions

Digital-religion scholarship asks **how religion is contested online**. That question is increasingly answered at a **scale** that manual reading alone cannot cover.

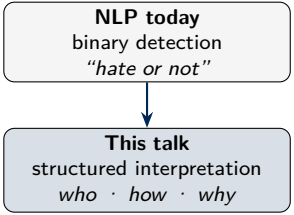
Most NLP work, by contrast, still asks only:

"is this post hate speech — yes / no?"

My question is different:

- **What kinds of anti-religious narratives emerge *across cultures***, once we stop forcing a binary label?

Bridge. This re-purposes AI as an **interpretive tool** for religion research — letting the data itself reveal the **forms** that religious hostility takes online.



A three-fold research gap

1. **Religion in NLP** $\approx$ Islamophobia + antisemitism.
   Anti-Christian hostility is comparatively under-studied.
2. **Languages**. Chinese and Japanese are rarely part of cross-lingual benchmarks; zero-subject grammar further breaks standard parsers.
3. **Task formulation**. Binary classification cannot tell us *who* is targeted, *how* the attack is framed, or *why* it is morally persuasive in a given community.

Takeaway

We need an *interpretable, cross-lingual* analytical framework — not a better hate-speech detector.

Why use anti-Christian discourse as the probe?

Christianity is analytically unusual because its social position **varies sharply** across the three communities we study:

- **English** — historically *dominant*, often critiqued from **within** a shared normative framework.
- **Chinese** — a *minority* faith frequently framed through **foreign influence**, missionary history, and state regulation.
- **Japanese** — a *small* minority, but politically salient after the Abe/Unification-Church scandal of 2022.

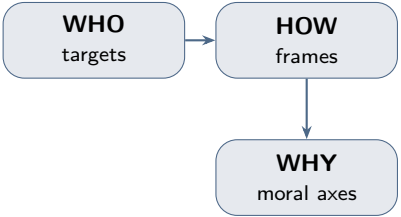
The same target, three distinct social positions: a natural laboratory for asking whether hate discourse is **universal** or **context-dependent**.

The goal is *not* to privilege one religion, but to use Christianity as a **comparative probe**.

Three linked questions

- **RQ1** — **Who** is targeted?
  Institutions, clergy, believers, public figures?
- **RQ2** — **How** is hostility expressed?
  Which rhetorical / thematic frames recur?
- **RQ3** — **Why** does it persuade locally?
  Which moral vocabularies carry the attack?

Together they form a reusable ***who–how–why* framework** for digital-religion analysis.



a reusable structured-interpretation framework

A trilingual corpus of 63,265 texts

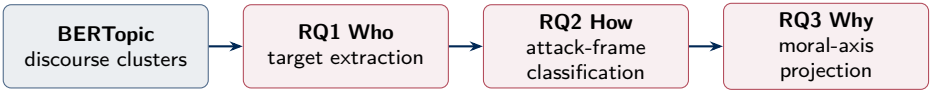
Language	Sources	Texts
English	Existing hate-speech datasets, curated forum threads	~21k
Chinese	Tieba + web corpora, custom annotation	~18k
Japanese	5ch + Common Crawl, custom annotation	~24k

- For low-resource settings, we use **LLM-assisted back-translation augmentation** rather than translating from English (which would import English framings).
- Annotation is **structured**: target, attack frame, and moral axis — not only "hate / not-hate".

Takeaway

A balanced trilingual resource lets us compare *framings* on equal footing — and is released open-source: `github.com/notsofun/Master_Thesis`.

Pipeline: from raw text to structured meaning



Design principle.

- **Induce** structure from the corpus, don't impose an English-trained taxonomy.
- **Modular** stages — each can be inspected, challenged, or replaced by a scholar.
- **Interpretable outputs** — a target label, a frame label, a moral vector.

Why not supervised end-to-end?

Pure supervised classifiers reproduce the labels we already have (mostly English, mostly Islamophobia / antisemitism). Unsupervised clustering lets new regions of the discourse *surface* themselves — including the Unification-Church cluster that nobody pre-labelled.

One technical slide — how the pieces fit

Shared backbone.
Multilingual E5-large (1024-d) encodes every sentence. A small mean-centring step per language aligns the three sub-spaces.

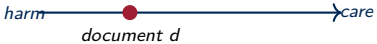
BERTopic (UMAP + HDBSCAN) then clusters the aligned embeddings into **47 discourse topics** — monolingual topics live next to genuinely cross-lingual ones.

For RQ2 we add a predicate-window extractor to cope with Chinese and Japanese zero-subject sentences, then an LLM classifier into **10 author-constructed attack frames**.

For RQ3 we build seven **moral axes** (5 MFT + 2 intergroup-threat) as centroid differences of dictionary words, and project every document onto them:

$$\mathrm{Bias}(d, k) \;=\; \cos\big(\mathrm{E5}(d),\, \mathbf{a}_k\big).$$

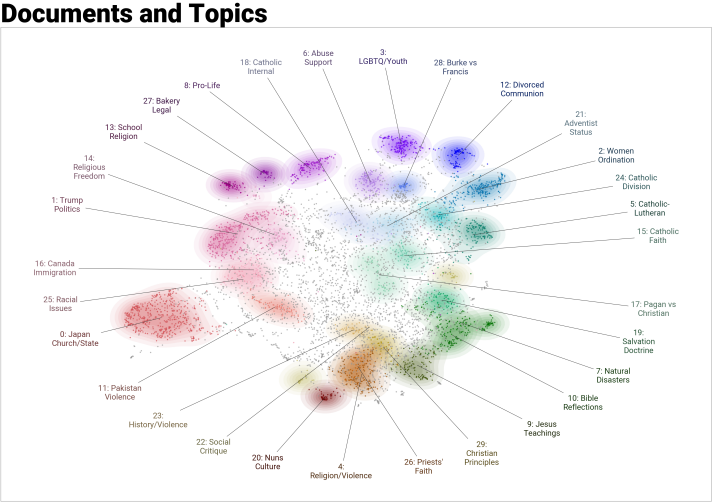
Axis = virtue centroid − vice centroid



A **negative** score means the text's embedding sits closer to the *vice/threat* pole — which is the fingerprint of hostile framing *via that moral axis*.

Crucially, a *positive* score can also be hostile when the speaker invokes ingroup virtues to attack an outgroup — more on this in a moment.

What the data says, before we ask any question



47 topics, coloured by dominant language.

Monolingual peaks.

- **JA (99%)** — Unification Church & Abe

Truly cross-lingual clusters. ($H \approx 1.1$)

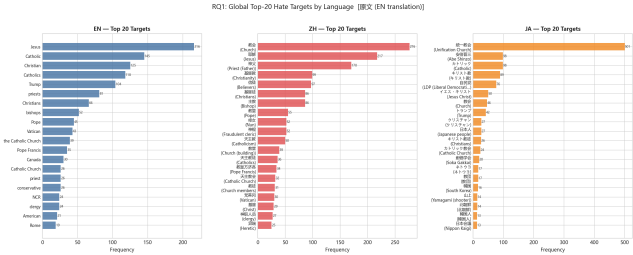
- LGBTQ+ & Catholic teaching

Finding 1

The target chooses the frame.

Online religious hostility follows reusable narrative templates that cross language boundaries.

RQ1 — Who is targeted?



Three distinct target profiles.

- **English** — *clergy* dominate; named political figures prominent (*priests, bishops, Trump*).
- **Chinese** — *church institutions* and the clerical hierarchy; plus "religious fraudster" — a register not seen in the other two languages.
- **Japanese** — (Unification Church) alone $= 501$ mentions, dwarfing every other target; paired with and .

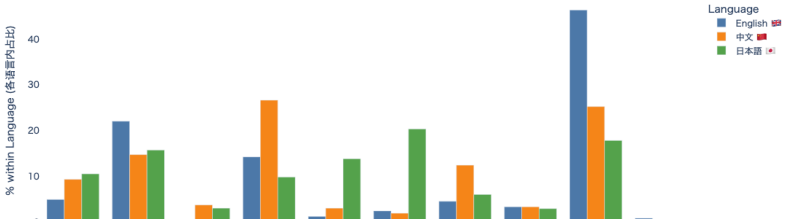
Japanese hate speech about religion is, in its most prevalent form, a discourse about **domestic political corruption**, not theology.

RQ2 — Which rhetorical frame carries the attack?

Ten author-constructed frames, induced from pilot-coding 200 documents and grounded in the framing literature:

- **Rhetorical Attack** — sarcasm, mockery, invective.
- **Moral Accusation** — hypocrisy, double standards.
- **Sexual Abuse Frame** — clerical abuse & cover-ups.
- **Economic Exploitation** — fraud, coerced donations.
- **Political Interference** — capture of state / party.
- **Cognitive Manipulation** — brainwashing, cult control.
- **Institutional Rot** — systemic cover-up.
- **Social Exclusion** — calls for banning / ostracism.
- **Dehumanization** — vermin / disease metaphors.
- **Other** — residual.

Fig B: Attack Framing Distribution by Language (% within language)



Three national “frame signatures”

English

- Rhetorical Attack *46%*
- Moral Accusation *22%*
- Sexual Abuse *14%*

“The phony, fake, stifling forced artificial piety.” — broad moral condemnation.

Japanese

- Political Interference *20%*
- Rhetorical Attack *18%*
- Economic Exploitation *14%*

Driven by Abe / : , , — politics and money.

Chinese

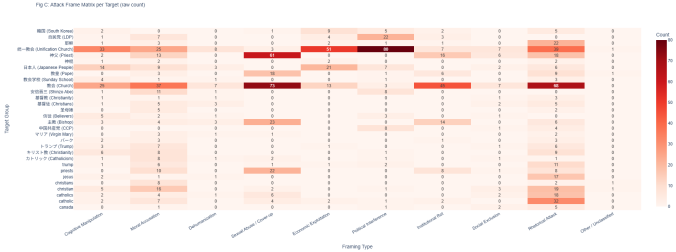
- Sexual Abuse *26%*
- Rhetorical Attack *25%*
- Moral Accusation *14%*

, , — clerical misconduct + cover-up.

Takeaway

Surface vocabularies differ; but — as the next slide shows — the *same underlying rule* chooses them.

The stable target–frame rule



Across ***all three languages*** *— not just within one — the target category selects the frame:*

Clergy (priests, bishops, Pope) ⟶ **Sexual Abuse Frame**

Institutions (Unification Church, LDP,) ⟶ **Political Interference / Economic Exploitation**

Collective believers (Catholics, ,) ⟶ **Rhetorical Attack / Moral Accusation**

Takeaway

Online hostility against religion behaves like a *genre* system: each social category of target activates a recognisable narrative template — **universal in form, localised in vocabulary**.

Finding 2

Two moral architectures, one shared target.

*English speakers attack Christianity as a **moral violator**.*
*Chinese and Japanese speakers attack it as a **threat to us**.*

How to read the moral profile

Each axis pairs a *virtue* pole with a *vice/threat* pole:

- **Harm** — care vs. harm
- **Fairness** — fairness vs. cheating
- **Loyalty** — loyalty vs. betrayal
- **Authority** — authority vs. subversion
- **Sanctity** — purity vs. degradation
- **Realistic Threat** — safety vs. invasion
- **Symbolic Threat** — cultural cohesion vs. cultural threat

The score is a cosine similarity between the document and the axis vector — it tells us *which moral vocabulary the document activates*, not whether the speaker is a fan of the axis.

Two hostile grammars both show up:

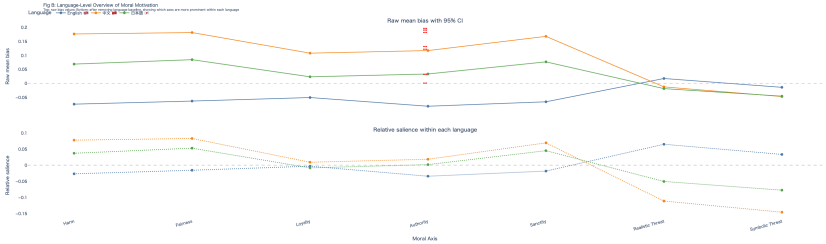
Direct condemnation

Vice words fall directly on the target.
"Christianity is corrupt, harmful, unjust."
⇒ **negative** MFT axes.

Identity defense

Ingroup virtue words defend "us"; target cast as threat.
"We must protect our sacred tradition from this foreign cult."
⇒ **positive** MFT + **negative** threat.

Three moral fingerprints



English — direct condemnation.
Negative on *all five* classical moral foundations. Christianity is indicted as "harmful, unfair, corrupt, authoritarian" — a **wide ethical indictment** from within a shared normative framework (insider critique).

Chinese & Japanese — identity defense.
Positive on the classical MFT axes; *negative* on Symbolic and Realistic Threat. Christianity is framed as a **foreign / social danger** — outsider critique organised around group safety.

The headline, in one sentence

"Christianity is morally wrong" *vs.* **"Christianity is dangerous to us."**

- Neither grammar is inherently more hostile than the other — they are *different rhetorical routes to the same end*.
- The choice is not random. It reflects each community's broader sense of what *a good society* should look like.
- *For Digital-Religion scholars:* the same target (Christianity) becomes an analytic lens onto the **moral imaginaries** of three online publics.

What the framework offers beyond this case

The *who–how–why* pipeline is **target-agnostic**. The same three-stage design can be pointed at:

- Islamophobia across Europe and South Asia
- Antisemitic rhetoric on Telegram, YouTube
- Anti-Hindu discourse in South-Asian diasporas
- Sectarian conflict (Sunni / Shi'a; Catholic / Protestant)
- Religion & nationalism studies
- Minority / majority reversals across regions

Takeaway

The contribution is a **framework**, not only a Christianity case-study — a reusable way for religion scholars to interrogate large-scale digital discourse at interpretable resolution.

Limits, and what I want to discuss with you

Honest limits.

- *Platform bias.* Tieba, 5ch, and Anglophone forums each carry their own selection effects.
- *Constructed datasets.* Our annotated Chinese / Japanese data are partly back-translation-augmented.
- *Moral dictionaries* were built for English; the CMFD / J-MFD adaptations are imperfect.
- *Qualitative validation* by religion scholars is still needed topic-by-topic — this is where **you** come in.

Three discussion questions.

1. How should Digital-Religion studies draw the line between *hostility* and *critique*?
2. Can computational models be trained to detect *theological* nuance — not only discursive hostility?
3. Which traditions should be compared next? Where would the *who–how–why* framework break?

Thank you.

From binary detection
toward **structured, cross-cultural interpretation**
of religious discourse online.

Zhidian Huang
huangzhidian15@gmail.com
github.com/notsofun/Master_Thesis

I'd love your questions — especially the hard ones.